

Lab-01 Student Club Event Ticketing & Budget Portal

FUNCTIONAL REQUIREMENTS (FR)

ID	Description	Priority	Acceptance Criteria	Rationale
FR-001	The system shall route event budget proposals through a multi-stage approval workflow	High	Status transitions sequentially upon digital sign-off; request cannot reach Dean without Finance Coordinator approval.	Ensures financial oversight and compliance with university budget policies.
FR-002	The system shall allow Club Leads to submit event proposals with details including event name, date, time, expected attendance, budget amount, and description.	High	All mandatory fields are validated and required before submission; submission fails if any field is missing.	Captures essential event information needed for review and budget allocation.
FR-003	The system shall generate unique, secure QR-code tickets for each attendee upon event approval and budget confirmation.	High	Generated QR codes are valid for single entry only; duplicate scans are rejected; QR codes contain event ID, attendee ID, and timestamp.	Enables secure venue entry and prevents duplicate attendance claims.
FR-004	The system shall provide Campus Admin with a dashboard to view all pending proposals, approval status, and generate event reports.	Medium	Dashboard displays real-time status of all proposals; filtering by status, club, and date range is functional; reports export to PDF format.	Improves administrative visibility and streamlines decision-making.
FR-005	The system shall allow registered attendees to download/view their QR-code tickets and provide venue check-in status notifications.	Medium	Attendees receive ticket link via email/portal within 1 hour of payment confirmation; check-in status updates within 5 minutes of venue scan.	Enhances user experience and provides real-time event attendance tracking.

NON-FUNCTIONAL REQUIREMENTS (NFR)

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The QR-code ticket scanner API shall validate entry authenticity in under 100 ms at venue check-in points, even under peak load (500+ simultaneous scans).	High	Benchmarking tests confirm latency $\leq 100\text{ms}$ under simulated 500+ concurrent requests; false acceptance rate $\leq 0.1\%$.	Ensures smooth venue entry flow without bottlenecks and prevents fraudulent ticket validation.
NFR-002	Usability & Availability	The system shall maintain 99.5% uptime during event submission and approval windows; UI shall be responsive across desktop, tablet, and mobile devices (load time $\leq 2\text{s}$).	High	Uptime monitored via SLA reporting; UI renders correctly on Chrome, Firefox, Safari, iOS Safari, and Android browsers; Lighthouse score ≥ 80 .	Ensures critical processes remain accessible to all stakeholders; accommodates diverse device usage patterns.

UML Use-Case Diagram

